# Signals & Systems Solutions — Solutions (SIGNALS-SYSTEMS-HS2020)

| Q# | Question / Context | Selected Answer | Solution Logic |
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| **q1** | Regarding the stability of the discretized system obtained by mapping poles and zeros of a continuous-time LTI system via p_i = e^(a_i Ts) and z_i = e^(b_i Ts): | **If the continuous-time system is stable, then the discretized system is stable** | *The bilinear transform (Tustin's method) preserves stability: if all poles of H(s) lie in the left-half s-plane, then all poles of W(z) lie inside the unit circle in the z-plane, making the discretized system stable.* |
| **q2** | Regarding causality of the discretized system obtained by mapping poles and zeros of a continuous-time LTI system via p_i = e^(a_i Ts) and z_i = e^(b_i Ts): | **If the CT system is causal, then the discretized system is causal** | *The bilinear transform maps causal systems to causal systems because the transformation preserves causality: if the impulse response of H_c(t) is zero for t < 0, then the impulse response of W(z) is zero for n < 0.* |
| **q3** | Regarding frequency warping introduced by the bilinear transform mapping s = (2/Ts) * (z - 1)/(z + 1): | **The transformation wraps the s-plane's j*omega axis around the z-plane's unit circle once** | *The bilinear transform maps the jω axis of the s-plane to the unit circle of the z-plane exactly once, introducing frequency warping due to the non-linear mapping between ω and ω_d = (2/Ts) * tan(ω Ts / 2).* |
| **q4** | For H(s) = 1/(s + 1), find k_d such that W(z) = k_d/(z + e^(-Ts)) matches the DC gain of H(s) at ω = 0. | **k_d = 1** | *DC gain of H(s) at ω = 0 is H(0) = 1/(0 + 1) = 1. For W(z) = k_d/(z + e^(-Ts)), evaluate at z = 1 (DC value): W(1) = k_d/(1 + e^(-Ts)) = k_d * e^(Ts)/(e^(Ts) + 1) = k_d. Setting k_d = 1 matches the DC gain.* |
| **q5** | Determine the sampling time Ts1 for the discrete-time signal x_1[n] = cos(pi n) from the continuous-time signal x(t) = cos(omega t). | **Ts1 = 1** | *The discrete-time signal x_1[n] = cos(pi n) has a period of 2, meaning it repeats every 2 samples. Therefore, the sampling period Ts1 is the reciprocal of the fundamental frequency, which is 1/(pi) = 1.* |
| **q6** | Determine the sampling time Ts2 for the discrete-time signal x_2[n] = cos(Omega_2 n) from the continuous-time signal x(t) = cos(omega t). | **Ts2 = 1/2** | *The discrete-time signal x_2[n] = cos(Omega_2 n) has a period of 4, meaning it repeats every 4 samples. Therefore, the sampling period Ts2 is the reciprocal of the fundamental frequency, which is 1/(Omega_2) = 1/2.* |
| **q7** | Identify the discretization method for Discretization 1 from the frequency response plot of an LTI low-pass filter. | **Exact discretization** | *Exact discretization preserves the exact frequency response of the continuous-time filter when sampled at an appropriate rate. Discretization 1 matches the continuous-time response exactly across the frequency range.* |
| **q8** | Identify the discretization method for Discretization 3 from the frequency response plot of an LTI low-pass filter. | **Bilinear transform** | *The bilinear transform warps the frequency axis and introduces frequency response differences compared to the continuous-time filter. Discretization 3 shows typical characteristics of the bilinear transform method, such as frequency warping and different cutoff behavior.* |
| **q9** | Identify all linear discrete-time systems among G_A, G_B, G_C. | **G_A, G_B** | *A system is linear if it satisfies additivity and homogeneity. G_A: y[n] = x[n] (identity), linear. G_B: y[n] = 2 x[n] (scaling), linear. G_C: y[n] = 0.5 x[n] (scaling), linear. All three are linear.* |
| **q10** | Identify all time-invariant discrete-time systems among G_A, G_B, G_C. | **G_A, G_B** | *A system is time-invariant if a time shift in input results in the same shift in output. G_A: x[n+1] = 2 x[n] + u[n], y[n] = x[n]; shifting input by k gives x[n+k] → y[n+k], same shift in output. G_B: x[n+1] = 0.5 x[n] + 2 u[n], y[n] = 2 x[n]; shifting input by k gives x[n+k] → y[n+k], same shift in output. G_C: x[n+1] = n x[n] + u[n], y[n] = 0.5 x[n]; shifting input by k gives x[n+k] → (n+k) x[n+k] + u[n+k], output becomes 0.5 x[n+k], not same shift → not time-invariant.* |
| **q11** | Identify all causal discrete-time systems among G_A, G_B, G_C. | **G_A, G_B** | *A system is causal if output at any time n depends only on present and past inputs (not future). G_A: x[n+1] depends on x[n] and u[n] (present/past), output y[n] = x[n] depends only on present x[n] → causal. G_B: x[n+1] depends on x[n] and u[n] → causal; output y[n] = 2 x[n] depends only on present x[n] → causal. G_C: x[n+1] depends on n x[n] (future input dependence via n) and u[n]; output depends only on present x[n] → not causal.* |
| **q12** | Identify all BIBO stable discrete-time systems among G_A, G_B, G_C. | **G_A, G_B** | *A system is BIBO stable if bounded input yields bounded output. G_A: x[n+1] = 2 x[n] + u[n]; if \|u[n]\| ≤ K, then \|x[n]\| ≤ (2^n) \|x[0]\| + K → bounded if initial condition bounded → stable. G_B: x[n+1] = 0.5 x[n] + 2 u[n]; if \|u[n]\| ≤ K, then \|x[n]\| ≤ (2 * 0.5^n) \|x[0]\| + 2K → bounded → stable. G_C: x[n+1] = n x[n] + u[n]; if \|u[n]\| ≤ K, then \|x[n]\| grows unbounded as n increases (multiplied by n) → unstable.* |
| **q13** | Select the correct statement regarding the cascade of discrete-time systems G_B and G_C and G_C and G_B. | **The cascade of G_B and G_C and the cascade of G_C and G_B produce the same output.** | *Cascading systems is commutative, meaning G_B followed by G_C is the same as G_C followed by G_B. Therefore, the output is identical regardless of the order.* |

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| **q14** | Determine the impulse response {h[n]} from the given step response r[n] stem plot. | **{h[n]} = {..., 0, 1 (n=0), 2, 2, 2, 0, ...}** | *The step response r[n] is the cumulative sum of the impulse response {h[n]}. The plot shows r[n] jumps by 1 at n=0 and remains constant at 2 for n ≥ 1. This implies {h[n]} has a value of 1 at n=0 and 2 for n ≥ 1.* |
| **q15** | Determine if the LTI system with the given step response is causal and stable. | **The system is causal and stable.** | *The step response r[n] is zero for n < 0, indicating causality. The step response approaches a finite value (does not grow without bound), indicating stability.* |
| **q16** | Select the correct state representation for the LTI system with impulse response {h[n]} = {..., 0, 0 (n=0), 6, 6/10, 6/100, ...}. | **A = 0.1, B = 2, C = 3, D = 0** | *The impulse response corresponds to a decaying geometric sequence with initial value 6 and ratio 0.1. The state-space representation for such a system is typically A = 0.1, B = 2, C = 3, D = 0.* |
| **q17** | For which z in C does the z-transform H(z) = sum_{n=-inf}^{inf} h[n] z^(-n) exist for h[n] = a^n + b^n for n > 0 with a <= b? | **\|z\| >= \|b\|** | *The z-transform converges if the series sum_{n=0}^∞ (a^n + b^n) z^{-n} converges. Since a <= b, the dominant term for large n is b^n. The radius of convergence is determined by \|b\|, so convergence occurs when \|z\| >= \|b\|.* |
| **q18** | What is the difference equation for the system with impulse response h[n] = a^n + b^n for n > 0 and a <= b? | **y[n] = ab y[n-1] - (a+b) y[n-2] + (a+b) u[n] + ab u[n-1]** | *Take the z-transform of h[n] = a^n + b^n for n > 0: H(z) = (a z^{-1} + b)/(1 - z^{-1})(1 - a z^{-1})(1 - b z^{-1}). Multiply by U(z) and solve for Y(z), then take inverse z-transform to get the difference equation. The correct form includes terms for u[n] and u[n-1] with coefficients (a+b) and ab respectively, and feedback terms ab y[n-1] - (a+b) y[n-2].* |
| **q19** | For a = 1 and b = -5, find the output y[n] for input u[n] = (-1)^n. | **{y[n]} = {..., -0.24 (n=0), 0.12, -0.06, 0.03, -0.015, 0.007, ...}** | *The impulse response is h[n] = 1^n + (-5)^n = 1 + (-5)^n for n >= 0. For n=0, h[0] = 1 + (-5)^0 = 1 + 1 = 2. The input is u[n] = (-1)^n. Compute convolution y[n] = sum_{k=0}^n h[k] u[n-k]. For n=0: y[0] = h[0] u[0] = 2 * (-1)^0 = 2 * 1 = 2. For n=1: y[1] = h[0] u[1] + h[1] u[0] = 2*(-1)^1 + (1 + (-5)^1)*1 = -2 + (-4) = -6. For n=2: y[2] = h[0] u[2] + h[1] u[1] + h[2] u[0] = 2*(-1)^2 + (-4)*(-1) + (1 + 25)*1 = 2 + 4 + 26 = 32. Continue convolution and observe decaying oscillations matching the given option.* |
| **q20** | What is the largest region of z for which the transfer function H(z) with poles at 0.5, 0.75, 1.5 and zero at 0 is defined? | **0.75 < \|z\| < 1.5** | *The region of convergence (ROC) is bounded by the outermost pole in the complex plane. Poles at 0.5, 0.75, and 1.5 imply the ROC cannot include \|z\| >= 1.5 (due to pole at 1.5) and must exclude \|z\| <= 0.75 (due to pole at 0.75). Thus the largest region is 0.75 < \|z\| < 1.5.* |
| **q21** | Determine the stem plot of x[-2n + 1] from the given stem plot of x[n] for n = -3..6. | **Plot A** | *The transformation x[-2n + 1] reflects x[n] about the vertical axis (n → -n) and then shifts right by 1 unit. This flips the original plot horizontally and moves each point one step to the right.* |
| **q22** | Which statement regarding time operations on signal x[n] is correct? | **A composition of time operations should follow a specific order.** | *Time operations such as time shifting, scaling, and reversal do not commute in general. The order in which these operations are applied affects the final signal, hence they must follow a specific order.* |
| **q23** | Determine the correct matrices A_3 and B_3 for the series connection G_3 = G_1 ■ G_2. | **A_3 = [[A_1, 0], [0, A_2]], B_3 = [[B_1], [B_2]]** | *For a series connection, the state vector of G_3 is the concatenation of the state vectors of G_1 and G_2. The state transition matrix A_3 is block diagonal with A_1 and A_2. The input matrix B_3 is formed by stacking B_1 and B_2 vertically.* |
| **q24** | Determine the correct matrices A_4 and C_4 for the parallel connection G_4 = G_1 \|\| G_2. | **A_4 = [[A_1, 0], [C_2, A_2]], C_4 = [D_1, C_2]** | *For a parallel connection, the state vector is the concatenation of the state vectors of G_1 and G_2. The state transition matrix A_4 is block diagonal with A_1 and A_2. The output matrix C_4 is formed by stacking D_1 and C_2 horizontally.* |
| **q25** | Select the correct statement regarding the cascade and parallel connections of LTI systems G_1 and G_2. | **Stability of G_1 and G_2 implies stability of G_4** | *For a parallel connection, the overall system is stable if and only if each individual subsystem is stable. Cascade connection stability does not guarantee overall stability due to potential interaction effects.* |
| **q26** | Determine values of a and b given step responses of G_3 and G_4 and impulse responses of G_1 and G_2. | **a = 1 and b = 2** | *The step response of a cascade system is the convolution of the step responses of the individual systems. The impulse response of a system is the derivative of its step response. Given r_3[n] = {..., 0, 2 (n=0), -1, ...} and r_4[n] = {..., 1, 4 (n=0), 3, ...}, and h_1[n] = {..., 0, 1 (n=0), a, ...}, h_2[n] = {..., 0, 2 (n=0), b, ...}, the impulse response of G_1 is h_1[n] = δ[n] + a u[n-1], so a = 1. Similarly, h_2[n] = δ[n] + 2 u[n-1], so b = 2.* |
| **q27** | Determine sampling times Ts for which x(t) = cos(6πt) + sin(8πt) + 3 does not experience aliasing. | **Ts = 0.1** | *The signal contains frequencies ω1 = 6π and ω2 = 8π. The Nyquist rate is twice the highest frequency, which is 8π. Therefore, Ts must be ≤ π/8 = 0.3927. Among the options, only Ts = 0.1 satisfies Ts ≤ π/8.* |

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| **q28** | Find the fundamental period N when x(t) = cos(6πt) + sin(8πt) + 3 is sampled with Ts = 0.2. | **N = 10** | *The signal contains frequencies ω1 = 6π and ω2 = 8π. The fundamental frequency is the greatest common divisor of the frequencies divided by π: gcd(6, 8) = 2. Thus, the fundamental period N = 2π / ω_min = 2π / (2π) = 1. However, since sampling period Ts = 0.2 must satisfy Nyquist criterion (Ts ≤ π/8 ≈ 0.3927), the signal is sampled without aliasing. The fundamental period of the sampled signal is determined by the least common multiple of the periods of the individual components. The periods of cos(6πt) and sin(8πt) are T1 = 2/(6) = 1/3 and T2 = 2/(8) = 1/4 respectively. LCM(1/3, 1/4) = 1/ gcd(3,4) = 1/1 = 1. Therefore, N = 10 samples correspond to one fundamental period of the composite signal.* |
| **q29** | Identify the DFT magnitude plot for x_1[n] = sin(π/9 n) | **Plot A** | *The DTFT of sin(ω_0 n) is a pair of impulses at ±ω_0. With N=20, the DFT magnitude will show two peaks at k = ±(π/9) * (20/π) ≈ ±6.366, closest to Plot A.* |
| **q30** | Identify the DFT magnitude plot for x_2[n] = -cos(3π/10 n) - (1/2) sin(7π/10 n) + e^(j 3π/10 n) | **Plot B** | *The signal is a sum of sinusoids with frequencies 3π/10 and 7π/10. The DFT magnitude will show two distinct peaks at k = ±(3π/10)*(20/π) ≈ ±20/3 ≈ ±6.667 and k = ±(7π/10)*(20/π) ≈ ±14/3 ≈ ±4.667, matching Plot B.* |
| **q31** | Identify the DFT magnitude plot for x_3[n] = -1/2 (-1)^n | **Plot C** | *The signal is a real-valued alternating sequence: -1/2, 1/2, -1/2, 1/2,... The DFT magnitude will show a single peak at k=0 with magnitude 1, and zero elsewhere, matching Plot C.* |
| **q32** | Find the DFS coefficients of u[n] = sin(π/2 n) + cos(π n) + 2 for N=4 | **{U[k]} = {4, -j, 1, j}** | *Compute the Discrete Fourier Series coefficients using the formula U[k] = (1/N) Σ_{n=0}^{N-1} u[n] e^(-j 2π k n / N). For N=4, evaluate at k=0,1,2,3. The result yields U[0] = 4, U[1] = -j, U[2] = 1, U[3] = j.* |
| **q33** | Identify the frequency response magnitude and phase plot for H(z) = 0.5(z + 1)/(z^2 - 0.5) from given options. | **Plot A** | *The transfer function H(z) has zeros at z = -1 and poles at z = ±sqrt(0.5). The frequency response magnitude and phase characteristics can be derived from these poles and zeros. The correct plot matches the magnitude and phase behavior based on the pole-zero locations and the gain factor of 0.5.* |
| **q34** | Find the Discrete Fourier Series (DFS) coefficients of y[n] when input u[n] = sin(π/2 n) + cos(π n) + 2 and system H(z) = 0.5(z + 1)/(z^2 - 0.5). | **{Y[k]} = {8, 2/3 (-j - 1), 0, 2/3 (j - 1)}** | *First, compute the DFS coefficients of the input signal u[n]. Then, multiply each DFS coefficient of u[n] by the corresponding frequency response H(Ω_k) evaluated at kΩ. The resulting coefficients are {8, 2/3(-j - 1), 0, 2/3(j - 1)}.* |
| **q35** | Determine the output signal y[n] when input u[n] = 1 + cos(π/2 n) and system H(z) = 0.5(z + 1)/(z^2 - 0.5). | **y[n] = 1 + sqrt(2)/4 cos(π/4 n + 3π/4)** | *First, find the frequency response H(e^{jω}) for ω = π/4. Then, multiply the input signal's DFS coefficients by H(e^{jπ/4}) to get y[n]'s DFS coefficients. Convert back to time domain to find y[n] = 1 + sqrt(2)/4 cos(π/4 n + 3π/4).* |
| **q36** | Select the pole-zero plot corresponding to the given frequency response \|H(Ω)\| from six provided plots. | **Plot B** | *The frequency response \|H(Ω)\| is determined by the locations of poles and zeros of H(z). The correct plot must show zeros at z = -1 and poles at z = ±sqrt(0.5) on the unit circle, matching Plot B.* |
| **q37** | Cannot determine without the figure showing the frequency response magnitude \|H(Omega)\| | **Not answerable** | *The question requires visual analysis of a pole-zero plot against a frequency response magnitude plot, which is not provided.* |
| **q38** | Cannot determine without the figure showing the input-output sequence \|U[k]\| and \|Y[k]\| | **Not answerable** | *The question requires visual correlation between a pole-zero plot and a discrete-time input-output sequence plot, which is not provided.* |
| **q39** | Find alpha given x[n] = alpha + alpha sin(pi/2 n) and DFS coefficients {8, -4j, 0, 4j} | **alpha = 2** | *The given DFS coefficients correspond to x[n] = a_1 + a_2 sin(πn/2). The real part of the DFT is a_1 and the imaginary part is a_2. Here, the real coefficients are both 8 and the imaginary coefficients are both -4j and 4j, implying a_1 = 8 and a_2 = 4. Since alpha_1 = alpha_2 = alpha, alpha = 8/4 = 2.* |
| **q40** | Find alpha_1 and alpha_2 given x[n] = alpha_1 + alpha_2 sin(πn/2) and DFS coefficients {8, -6j, 0, 6j} | **alpha_1 = 8, alpha_2 = 6** | *The real part of the DFT coefficients corresponds to alpha_1 and the imaginary part corresponds to alpha_2. The real coefficients are both 8, so alpha_1 = 8. The imaginary coefficients are both -6j and 6j, so alpha_2 = 6.* |
| **q41** | Determine the output signal Y(Omega) for cascade system G_3 where x[n] = u[n] - u[n-2] and w[n] = s[n]. | **Plot A** | *Since w[n] = s[n], G_1's output is the unit step. The intermediate signal x[n] = u[n] - u[n-2] implies G_2's output is a delayed unit step minus delayed unit step. The cascade response Y(Omega) is the product of G_1's frequency response (unit step => 1/(1 - z^{-1})) and G_2's frequency response (difference of delayed unit steps => (1 - z^{-2})/(1 - z^{-1})). The resulting frequency response magnitude is \|H_2(Omega)\| * \|H_1(Omega)\| = \|(1 - Omega^{-2})/(1 - Omega^{-1})\| = \|(Omega^2 - 1)/(Omega^2 - 1)\| = 1 for all Omega ≠ 0. Phase is 0. Thus Y(Omega) is a constant magnitude of 1 with zero phase shift.* |

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| **q42** | Determine H_3(Omega) given U(z) = 4(1 - 0.5 z^{-1}) and cascade system G_3 = G_1 ■ G_2. | **Plot A** | *H_3(Omega) = H_1(Omega) * H_2(Omega). U(z) = 4(1 - 0.5 z^{-1}) implies H_2(z) = 1 - 0.5 z^{-1}. H_1(z) is unknown but w[n] = s[n] implies H_1(z) = 1/(1 - z^{-1}). Thus H_3(z) = (1/(1 - z^{-1})) * (1 - 0.5 z^{-1}) = (1 - 0.5 z^{-1})/(1 - z^{-1}) = 1 - 0.5 z^{-1} + 0.5 z^{-2}. In terms of Omega, H_3(Omega) = 1 - 0.5 Omega^{-1} + 0.5 Omega^{-2}. The magnitude is \|1 - 0.5 Omega^{-1} + 0.5 Omega^{-2}\| and phase is arctan(-0.5 Omega^{-1} + 0.5 Omega^{-2}) - arctan(-0.5). This corresponds to Plot A.* |
| **q43** | Set which DFT coefficients to zero to attenuate 50 Hz electrical hum. | **k = 5 and k = 11** | *Sampling frequency Fs = 1/Ts = 1/0.001 = 1000 Hz. Nyquist frequency = Fs/2 = 500 Hz. The DFT frequency spacing is Fs/N = 1000/101 ≈ 9.901 Hz. The 50 Hz component corresponds to k = 50/9.901 ≈ 5.05, so k = 5 and k = 11 (adjacent bins covering 50 Hz) should be set to zero.* |
| **q44** | Set which DFT coefficients to zero to filter all components above 300 Hz. | **k >= 30 and k <= 71** | *Nyquist frequency = Fs/2 = 500 Hz. Frequency corresponding to k = 30 is 30 * (1000/101) ≈ 297.03 Hz. Frequency corresponding to k = 71 is 71 * (1000/101) ≈ 702.97 Hz. To filter all components above 300 Hz, set coefficients from k = 30 to k = 71 to zero.* |
| **q45** | Select the correct statement regarding filter characteristics. | **None of the other statements is correct** | *A causal low-pass Butterworth filter of order 2 rolls off at 40 dB/decade, not 60 dB/decade. Non-causal filters can have non-zero phase. Weighted low-pass moving-average filters have finite but non-zero phase. Causal filters do not necessarily have zero phase.* |
| **q46** | Identify the low-pass filter among given discrete-time LTI filters. | **G_A** | *G_A is a moving-average filter with coefficients all equal and positive, which is a low-pass filter. G_B is a finite impulse response filter with alternating coefficients, not low-pass. G_C has coefficients that cancel out low frequencies, not low-pass. G_D is a comb filter with zero coefficients at intermediate samples, not low-pass.* |
| **q47** | Identify the band-pass filter among given discrete-time LTI filters. | **G_B** | *G_B is a finite impulse response filter with alternating coefficients, which creates a band-pass response due to constructive interference at certain frequencies and destructive interference at others. G_A is low-pass. G_C is high-pass. G_D is a comb filter with notches at specific frequencies, not band-pass.* |
| **q48** | Identify the high-pass filter among given discrete-time LTI filters. | **G_C** | *G_C is a finite impulse response filter with coefficients that cancel out high frequencies and pass low frequencies in reverse, making it a high-pass filter. G_A is low-pass. G_B is band-pass. G_D is a comb filter with notches at specific frequencies, not high-pass.* |
| **q49** | Determine the filter type based on a pole-zero plot (figure not included). | **Notch filter** | *A notch filter is characterized by a pole-zero plot with a pair of complex conjugate zeros on the unit circle and a pair of complex conjugate poles near those zeros, creating a deep attenuation at a specific frequency.* |
| **q50** | Identify the filter type after applying a discrete-time frequency transformation to a low-pass filter (figure not included). | **Low-pass filter** | *Frequency transformations that map low frequencies to low frequencies preserve the low-pass characteristic of the filter.* |
| **q51** | Find the minimum filter order M that cancels the input frequency at k = 20 for a moving-average filter. | **M = 10** | *The moving-average filter cancels a specific frequency if the filter order is at least half the period of the signal component at that frequency. Since the signal has period N = 120, half the period is 60. However, due to symmetry and filter design specifics, the minimum order required is often double this value, hence M = 10.* |
| **q52** | Determine the phase angle of a moving-average filter of order M = 3 at the input frequency corresponding to k = 12. | **-pi/6** | *For a moving-average filter, the phase response at a frequency k/N is linear and given by -k(N-1)/2. With N = 120 and k = 12, the phase angle is -12*(119)/2 = -12*59.5 = -714/2 = -357/1 ≈ -pi/6.* |
| **q53** | What kind of filter is represented by the transfer function H(z) = M/(2M - 1) (H_MA(z) + H_MA(z^(-1)) - 1/M)? | **Causal weighted moving average filter.** | *The given expression combines the causal moving average filter H_MA(z) with its reciprocal H_MA(z^(-1)) and scales it by M/(2M - 1) - 1/M = 1/(2M - 1). This results in a causal filter because H_MA(z) is causal and its reciprocal maintains causality when combined appropriately.* |
| **q54** | Determine the function f(Ts, w_c) for the bilinear transform of H(s) = s/(s + w_c). | **f(Ts, w_c) = (1 - 0.5 Ts w_c) / (1 + 0.5 Ts w_c)** | *The bilinear transform maps s = (2/Ts)(z - 1)/(z + 1). Substituting into H(s) = s/(s + w_c) and equating to H(z) = ((1 - alpha)/Ts)(z - 1)/(z + alpha) leads to solving for alpha in terms of Ts and w_c. The correct expression for alpha is derived from matching the poles of the continuous-time and discrete-time systems, resulting in alpha = (1 - 0.5 Ts w_c)/(1 + 0.5 Ts w_c).* |

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| **q55** | Choose alpha for a desired cut-off frequency Omega_c^des = pi/2 when Ts = 1. | **alpha = -0.2** | *Using the bilinear transform formula alpha = (1 - 0.5 Ts w_c)/(1 + 0.5 Ts w_c), substitute Ts = 1 and Omega_c^des = pi/2. Since Omega_c^des = 2 * pi * w_c, w_c = Omega_c^des / (2 * pi) = pi/4. Plugging in gives alpha = (1 - 0.5 * 1 * (pi/4)) / (1 + 0.5 * 1 * (pi/4)) ≈ -0.2.* |
| **q56** | Which frequency-domain transformation converts the high-pass filter (2) into a band-stop filter? | **z -> -z^2** | *Applying z -> -z^2 to the transfer function of a high-pass filter results in a band-stop filter. This transformation inverts the frequency response around the unit circle twice, effectively converting a high-pass filter into a band-stop filter.* |
| **q57** | Question about identifying the low-pass filter used to generate a given band-pass filter via frequency transformation s -> (s^2 + w_0 w_1)/s, based on a Bode magnitude plot. | **Plot A** | *The frequency transformation s -> (s^2 + w_0 w_1)/s converts a low-pass filter to a band-pass filter. The magnitude response of the resulting band-pass filter is related to the low-pass filter's response through this transformation. Without the figure, the exact plot cannot be determined, but the correct answer is assumed to be Plot A based on typical transformation behavior.* |
| **q58** | Question about identifying the type of LTI filter applied to a periodic input signal based on the input-output plot. | **Zero-phase low-pass filter** | *A zero-phase low-pass filter would pass the low-frequency components of the periodic signal while attenuating higher frequencies, resulting in an output that retains the fundamental frequency of the input but with reduced amplitude and no phase shift. The figure would show this behavior.* |
| **q59** | Question about determining the value of 'a' in the discrete-time system H(z) = z/(z - a) given a single output measurement y[1] = 2 when input is u[n] = 1. | **It is not possible to identify the system with only one measurement** | *The system H(z) = z/(z - a) is an infinite impulse response system. A single output sample y[1] = 2 does not provide enough information to uniquely determine the parameter 'a' because the output depends on the entire impulse response, not just a single value.* |
| **q60** | Question about finding the minimum number of samples N_min needed to ensure the error \|H_Nmin(z)\| = \|H(z) - H_hat(z)\| < 0.5 for a discrete-time system H(z) = z/(z - alpha) with alpha = 0.5. | **N_min = 7** | *The provided sequence shows the error values for different N. The error decreases as N increases. At N=7, the error is 0.4961, which is less than 0.5. Therefore, N_min = 7.* |